Network Protocol

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1. Encoding and Decoding

Every network protocol message sent in the network begins with a four capital letter sequence which matches one of the predefined network commands, presented later in this document. Depending on the command itself, it can be followed by any number of arguments. In case it is necessary an argument can be encoded by simply preceding one digit in front of it, which relays the length of the argument together with this one encoded digit. As long as the amount of arguments is set, this can be expanded to more than one argument. An abstract representation of a network message:

<4 letter protocol><digit><1. argument><digit><2. argument>…<digit><n. argument><msg>

A received network protocol message can be decoded as long as the amount of arguments is known.

2. Network Protocol Commands

LGIN - Login

This command signals that a new player / client has connected to the server.

LOGT - Logout

This command signals that a player wants to logout from the server and leave the game. The server closes the connection to the client and updates the client list. No data string has to be attached.

QUIT - Quitting the game

This command transmits that a player wants to leave the game. The server makes sure to close the connection and update the player list. No data string has to be attached.

NAME - Set Username

This command signals the player's intention to change their username. The server checks if the username is already being used by some other client and changes it if needed. The attached data string contains the desired new username. This command is only sent from clients to the server.

HSCR - Highscores

This command shows the player's intention to have a look at the high score list. The server transmits the high score list to the client. If the command is sent from the client to the server, then no data string is required.

MSGE - Messages

This command sends responses from the server to the clients in the form of messages. The data string includes the message.

CELY - Create new a Lobby

This command signals that the client wants to create a new lobby. The server creates the new lobby and adds the client to it. This command is sent from the client to the server and no data string is needed.

JOLY - Join Lobby

This command indicates that a player wants to join an already existing lobby. The name of the lobby is attached in the data string. This command is only sent from client side to server side.

QILY - Leave Lobby

This command signals the player's intention to leave his lobby. The server displays a leaving message and removes the client from the lobby and the player list. This command is only sent from clients to the server.

NEWG - New Game

This command signals the player's intention to start a new game. The server now initiates a new game with the client's lobby.

MOVE - Move

This command sends the movement of a chess figure. The data is formatted as follows:
"Starting Position" + "Ending Position" and encoded by the encoder.

CHBD - Chessboard

This command is used to update the chessboard after every move. The server then updates the board and returns it to the players .

SELT - Select

This message signals that a player has selected a specific figure on the board. This command is sent from the client to the server.

SLTD - Selected

This command tells all clients which figure has been selected by the current player. This command is sent from the server to the clients.

TURN - Turn

This command tells all the players that a new turn has begun and who's turn it is. The current player's ID is transmitted as the data.

STAT - Status

This command relays the current status of a lobby chess board. It is sent from server to client.

GOVR - Game Over

This command signals that the game is over. The command is sent from the server to the clients and the attached string contains the information about the game result.

ENDG - End the game

This command is used to end the game. The server calculates which team has more points and returns the result to the players.

DRAW - Game ends in a draw

This command is used when the game ends in a draw. The server returns a message with the points that were achieved by both teams.

WINN - Winner

This command returns the winner of the game after the game has ended. It is sent from the server to the clients. The data string returns the winners and the achieved points.

UPPL - Update Players

This command is used to update the list that contains all players. The command is sent from the server to the clients.

CHAT - Global Chat

This command signals that a client wants to send a message to all the connected clients. The server then transmits this message to all the clients that are in the global chat.

WCHT - Whisper Chat

This command signals that a client wants to send a private message to another client. This chat message is sent from a client to a server and then redirected to the recipient.

LCHT - Lobby Chat

This command signals that a client has sent a message in the lobby chat. The server sends the message only to the players who are connected to the same lobby. The data string contains the message.

SLOB - Show Lobbies

This command shows all available lobbies to the player. The server sends a list of all lobbies as a data string encoded by the encoder.

LBBS - Update the lobbies

This command is used to update the list that contains all lobbies. It is sent from the server to the clients.

SNAM - Show Name

This command is sent from the client to the server. The server then returns the client's username.

SLNA - Show Lobby Name

This command sends the name of the lobby the client currently is in, to the client as a data string. This command is sent from the client to the server.

SPLY - Show players

This command shows the client how many players are already in the lobby. It transmits a simple integer as data.

SHGM - Show Games

This command indicates that the player wants to see a list of all the games that are currently being played. The server returns a data string containing all the ongoing games.

PSMS - Past Games

This command indicates that the player wants to see a list of all finished games. The server then returns a list with all the finished games.

OPGM - Open Games

This command indicates that the player wants to see a list of all open games ready to be joined. The server then returns a list with all the lobbies that are still waiting for players.

TMER - Timer

This command starts a timer after initiating a game. It is sent from the server to the clients.

PING - Ping

This command is used to ping the server and check whether the connection is still standing.

PONG - Pong

This command is used to pong the client and check whether the connection is still standing.

SONO - Sound

This command is used to tell the client when and which sound files to play. It is sent from the server side based on the current game stance. The data part of the message contains a string that tells the client which sound to play, for example:
“Start”
“Piece captured”
“Check”